Space Center Houston
Conrad Challenge
Presented by equinor

The Lean Canvas Model

Company Name: Nazarbayev Intelllectual school of Astana

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Problem

Every year, 1.4 million tonnes of plastic flow from rivers into oceans, while poor monitoring leaves pollution unnoticed, threatening aquatic ecosystems and making it harder to achieve UN Sustainable Development Goals 6 (Clean Water) and 14 (Life Below Water)

6 CLEAN WATER AND SANITATION

14 LIFE BELOW WATER

Existing Alternatives

Ocean Cleanup Interceptor: Industrial river cleanup costing hundreds of thousands of dollars.
RanMarine WasteShark: Autonomous waste collection with optional water-quality monitoring.
WaterGuard: Integrates AI detection, water monitoring, and autonomous cleanup in one solar-powered system for approximately $397.

Solution

WaterGuard station is a closed-loop system that integrates continuous **Computer-vision** object detection, real-time water **quality sensing** and a self-propelled, automatic waste-collection **mini-robot.** It is self-powered and works **24/7** in isolation with considerably lower cost than present alternatives.

Key Metrics

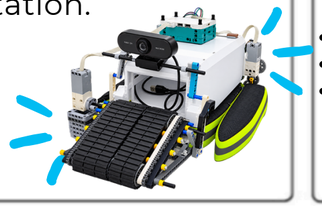
- Robot testing completed in real water bodies.
- Number of successful field trials.
- Amount of commercials deal (One is already currently under negotiation).
- Mass of waste that was collected in a hour.
- Percentage successful readings of robot's sensors.

Unique Value Proposition

Our innovative system integrates 24/7 fully autonomous AI detection with a dedicated cleanup robot and a fully integrated web platform. It is automatic, unlike costly and manually operated alternatives, to gather floating debris and provides real-time water analytics at a much lower cost.

High-Level Concept

Autonomous AI-powdered water monitoring and waste collection station.



Sustainable Advantage

AI

- Cost of **$394** is unmatched, making it a cost that is ***5 to 20 times cheaper*** than options in the market **($1,800–$7,000).**
- Platform that integrates **Closed-Loop Automation, IoT water-quality sensing, and robotic cleanup**, and incorporates **edge AI vision (*YOLO*).**

Channels

- Engaging **directly with government agencies**, water management departments, and park authorities through public procurement as B2G & Municipal Tenders.
- **Expos and Eco-Events**
- **Pilot Deployments**
- Through **digital channels and social media**, B2C businesses can be engaged and awareness is promoted.

Customer Segments

- **B2B:** Reservoir operators and environmental companies that need monitoring and floating waste removal systems.
- **B2C:** Pond owners seeking for water quality monitoring services and reliable data about reservoirs.
- **B2G:** Water departments and environmental agencies.

Early Adopters

- Syr Darya–Turkestan State Regional Nature Park - initial testing and potential deployment.
- Other natural reserves and national parks in Kazakhstan.
- Local environmental authorities managing polluted waterways.
- River conservation and water management organizations.









Cost Structure

The price of a Raspberry Pi 5 and ESP32 is 103,000 KZT.
Sensors, GPS and camera: 32,000 KZT.
Solar power and batteries: 27,000 KZT.
The cost of motors, electronics, and structural components is 18,000 KZT.
The robot-station system's total production cost is 180,000 KZT.

Revenue Streams

Revenue streams will come from Saas subscriptions for real-time data access and cloud analytics, hardware station and cleaning robot sales, and recurrent maintenance, servicing and government bids for municipal water management.

